

Assignment 6: Secure Data/Computation Outsourcing

COMS/CPRE 4590X/5590 Spring 2026

Due: Wednesday, April 22

Total Credits: 16 points

Instructions: Some questions may have multiple correct answers.

Part 1: Attribute-Based Encryption (ABE)

Q1. Carol wants to store a file in the cloud and allow access only to users who satisfy all of the following: [2 pts]

- Age ≥ 21
- Department = Electrical Engineering
- Completed course EE 301

Can Carol use only ABE to enforce this policy?

- If no, briefly explain why.
- If yes:
 - What attributes are needed?
 - What is the access structure tree?

Q2. Dave wants to allow access if: [2 pts]

- User is a Manager, OR
- User is an HR employee, OR
- User is an Intern AND the file has already been accessed by a Manager

Can ABE alone enforce this policy? Briefly explain.

Q3. Which of the following are true for ABE in cloud storage? [0.5 pts]

- (a) The cloud must decrypt data
- (b) The cloud enforces access
- (c) The data owner can go offline
- (d) Users must authenticate

Q4. Which are required for ABE encryption? [0.5 pts]

- (a) Plaintext document
- (b) Master secret key
- (c) Public parameters
- (d) Access policy

Q5. To successfully decrypt a given ciphertext, which of the following are required? [0.5 pts]

- (a) Access policy
- (b) Master secret key
- (c) Public parameters
- (d) Attribute set satisfying policy

Part 2: Oblivious RAM (ORAM)

Q6. ORAM protects which of the following? [0.5 pts]

- (a) Access frequency
- (b) Access order
- (c) Timing of accesses
- (d) Read vs. write type

Q7. In Path-ORAM, what must the client store? [0.5 pts]

- (a) Encryption key
- (b) Overflow blocks
- (c) Exact node locations
- (d) Leaf/path assignment

Q8. Which are true about the server storage tree? [0.5 pts]

- (a) Blocks have equal size
- (b) Nodes must be fully filled
- (c) Dummy blocks may exist
- (d) Server knows dummy blocks

Q9. In a data access, the client must: [0.5 pts]

- (a) Download a full path
- (b) Decrypt all downloaded blocks
- (c) Re-encrypt blocks
- (d) Upload real blocks

Part 3: Garbled Circuits

Q10. Garbled circuits achieve which of the following? [0.5 pts]

- (a) Hide inputs from evaluator
- (b) Hide outputs from evaluator

- (c) Hide circuit size
- (d) Verify correctness

Q11. A circuit has 5 input wires, 4 internal wires, and 2 output wires. How many keys are needed? [0.5 pts]

- (a) 10
- (b) 11
- (c) 18
- (d) 22

Q12. When garbling a gate, which are required? [0.5 pts]

- (a) Two keys per input wire
- (b) Two keys per output wire
- (c) Encrypt truth table rows
- (d) Randomly permute rows

Q13. The evaluator must receive: [0.5 pts]

- (a) One key per input wire
- (b) Two keys per input wire
- (c) One key per output wire
- (d) Two keys per output wire

Part 4: Searchable Encryption

Q14. The final scheme requires: [0.5 pts]

- (a) Encryption scheme
- (b) Digital signatures
- (c) Pseudorandom number generator
- (d) None of the above

Q15. Which statements are true for their final scheme? [0.5 pts]

- (a) Keyword is sent in plaintext
- (b) Keyword is encrypted
- (c) Server learns all positions where the keyword occurs
- (d) Client can decrypt the document

Part 5: Multi-Party Computation

Q16. For 1-out-of-2 Oblivious Transfer: [0.5 pts]

- (a) Sender encrypts both messages
- (b) Sender knows receiver's choice
- (c) Receiver obtains both messages

- (d) Receiver obtains only one message

Q17. Let Alice hold $a_1 = 0, b_1 = 1$ and Bob hold $a_2 = 1, b_2 = 1$. Let $c = a \times b$. Possible output shares (c_1, c_2) are: [1 pts]

- (a) (0,0)
- (b) (0,1)
- (c) (1,0)
- (d) (1,1)

Part 6: Shamir Secret Sharing

Q18. Secret $s = 5$ is distributed as Shamir shares, using polynomial $f(x) = 5 + 3x$ over $\mathbb{Z}_{11}$. Number of shares $n=3$ and threshold $t=1$. [1 pts]

- Compute all shares
- Reconstruct using two shares

Q19. Which are true? [0.5 pts]

- (a) Any $t + 1$ shares reconstruct
- (b) Fewer than t reveal nothing
- (c) Uses random polynomial
- (d) Secret must be zero

Q20. Polynomial degree is: [0.5 pts]

- (a) n
- (b) $t - 1$
- (c) t
- (d) Always 1

Part 7: MPC with Secret Sharing

Q21. Over $\mathbb{Z}_7$, given $a_1 = 2, a_2 = 3$ and $b_1 = 4, b_2 = 1$: [1 pts]

- Compute shares of $c = a + b$
- Reconstruct c

Q22. Which operations are local (no interaction)? [1 pts]

- (a) Addition
- (b) Multiplication
- (c) Scalar multiplication
- (d) Reconstruction